Electronics and Computer Science

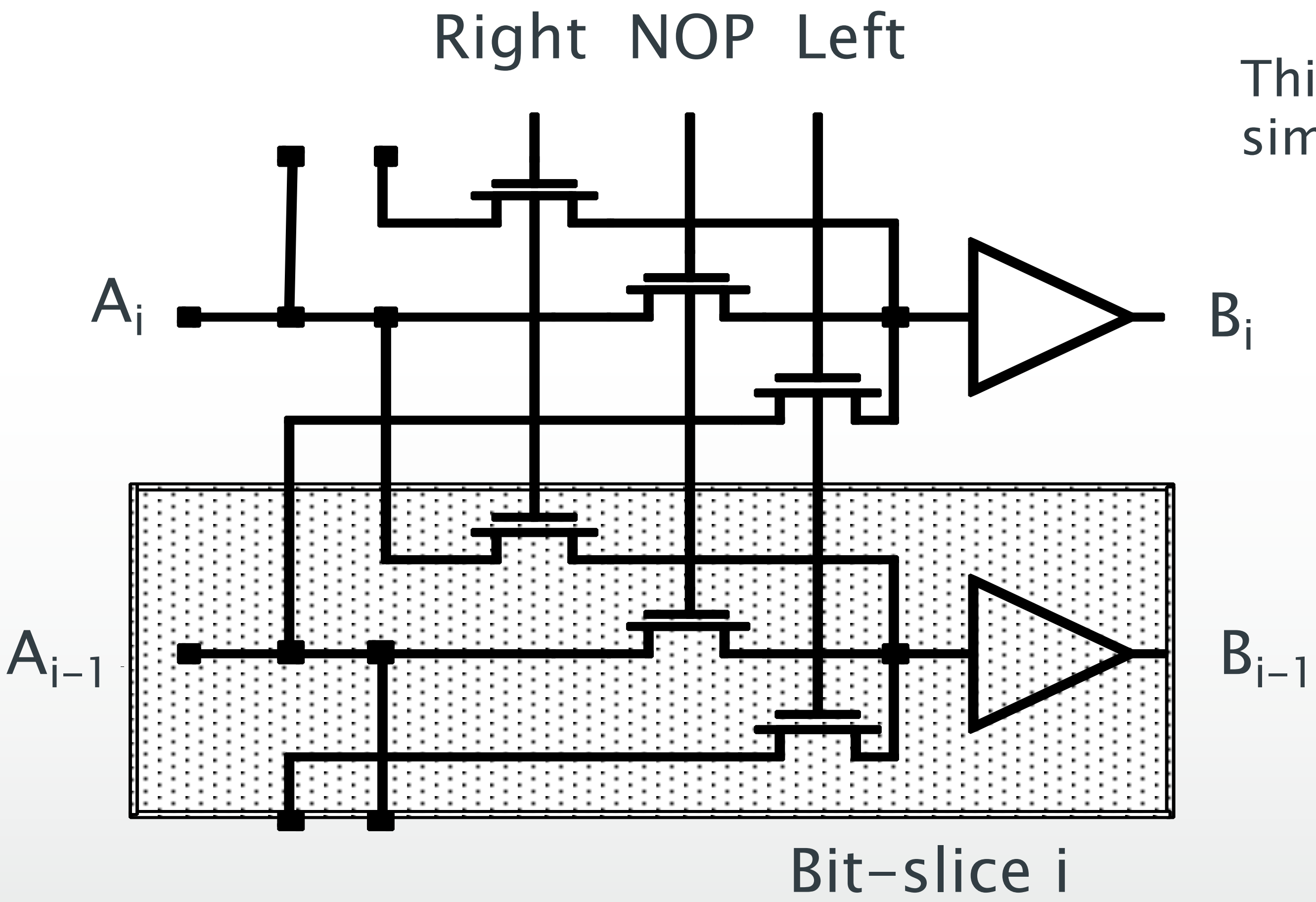
UNIVERSITY OF Southampton

15: Computer Arithmetic – Shifting and complex arithmetic

Shift and Rotate operation

- Used in microprocessor and encryption algorithm
    - Shift left by one place = multiply by 2
- If fixed shift: simply wire the inputs to the correct output position
- Variable shift
    - One bit shifter
    - Barrel shifter
    - Logarithmic shifter

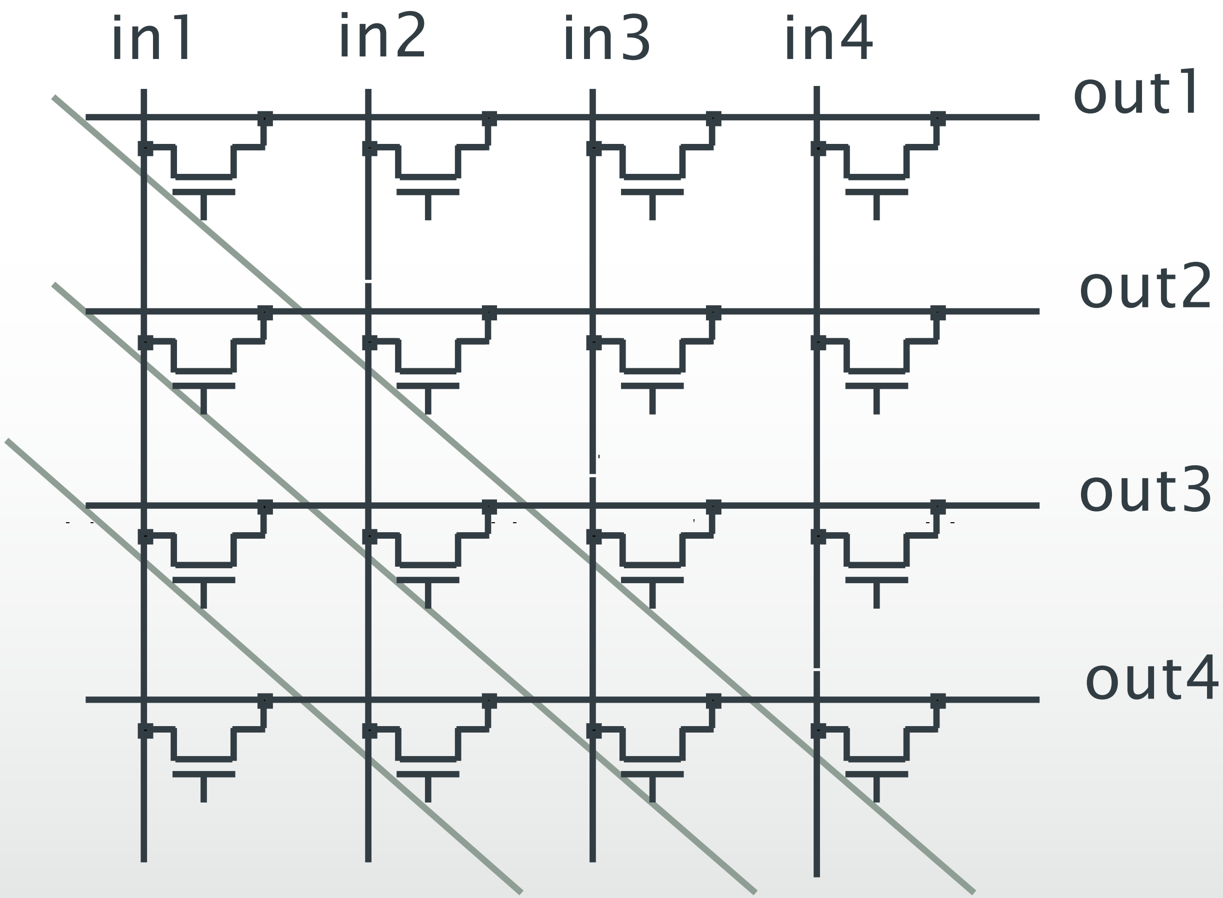
One bit shifter



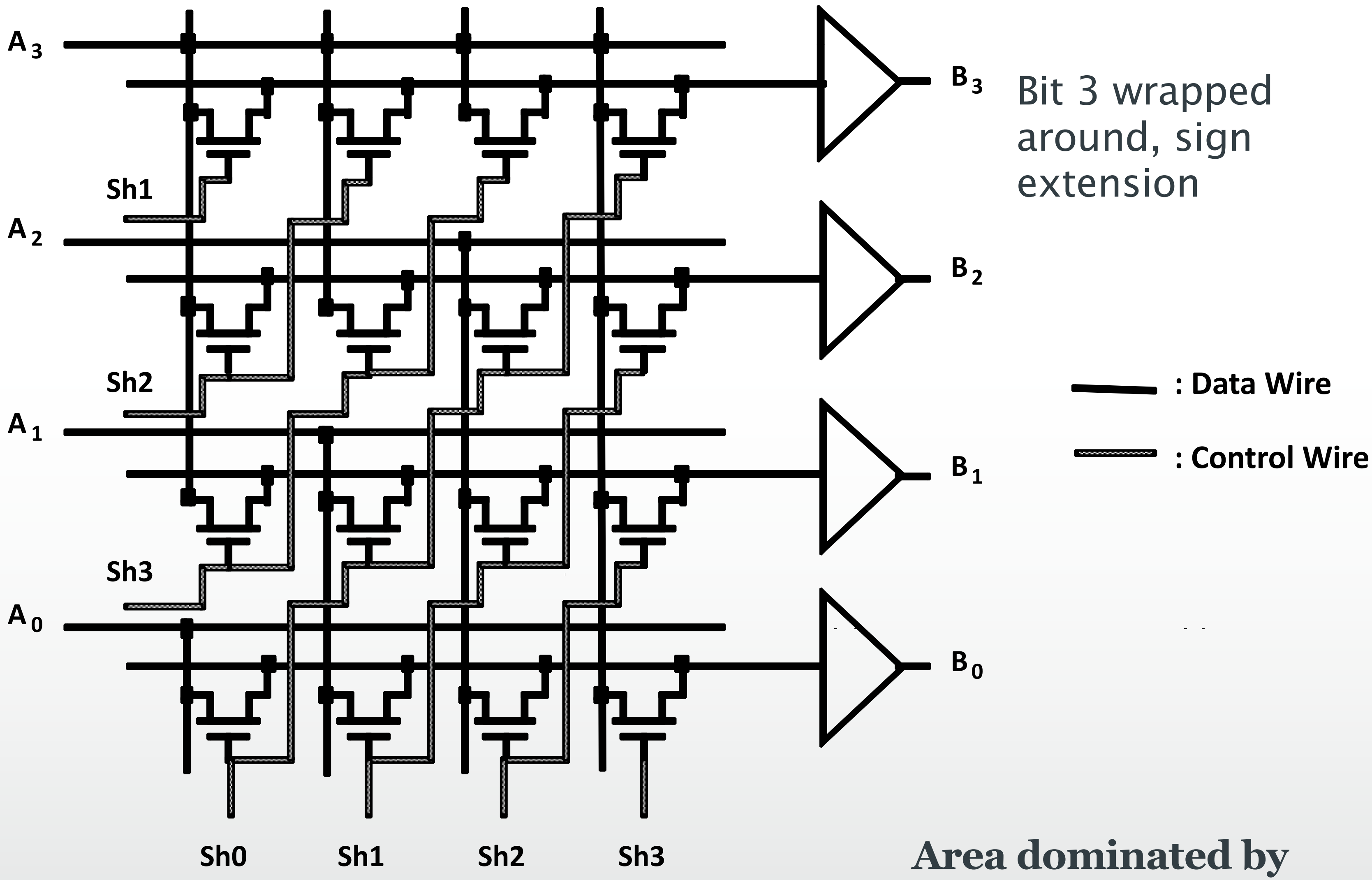
Think of transistors simply as switches

Simple N-bit shifter

- Quadratic number of transistors: one transistor switch per path



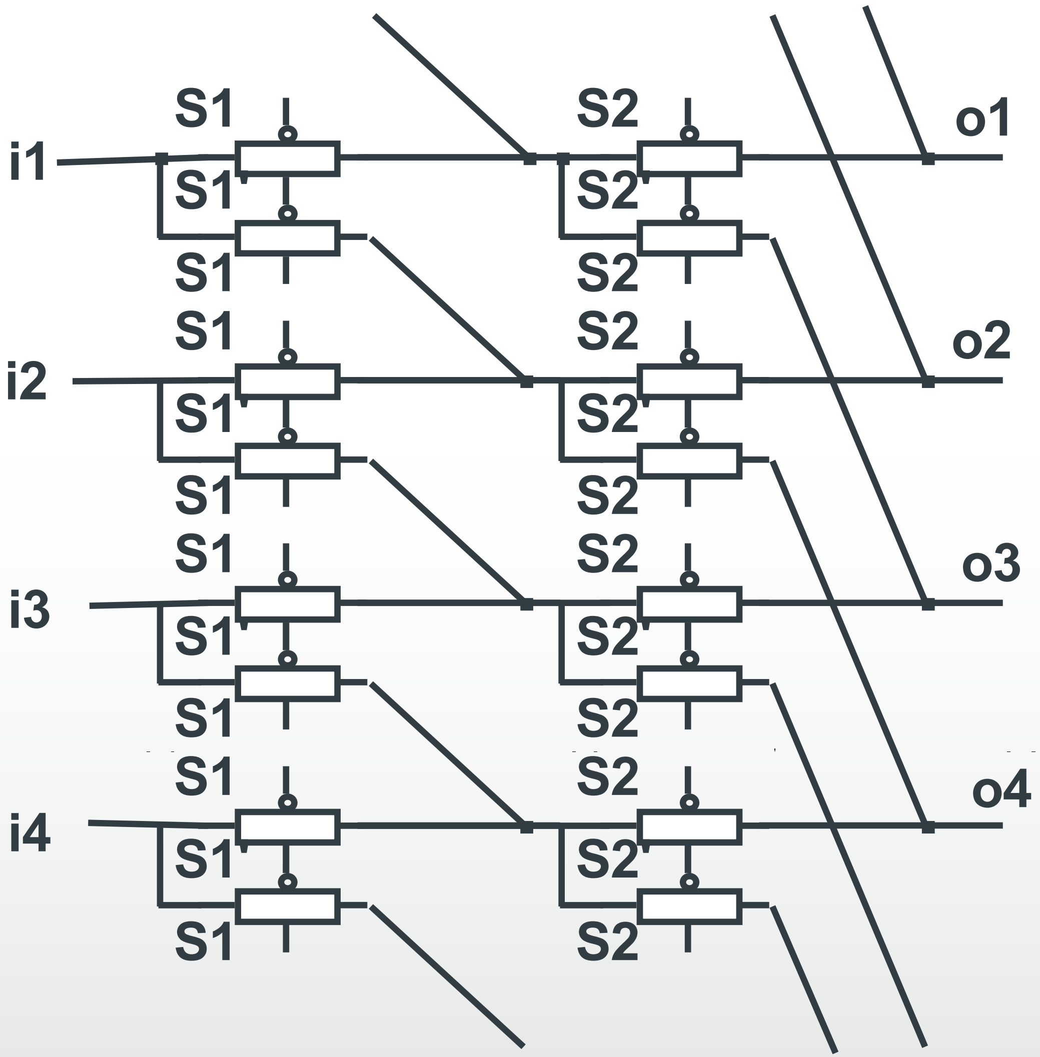
Barrel shifter



Bit 3 wrapped around, sign extension

Area dominated by wiring

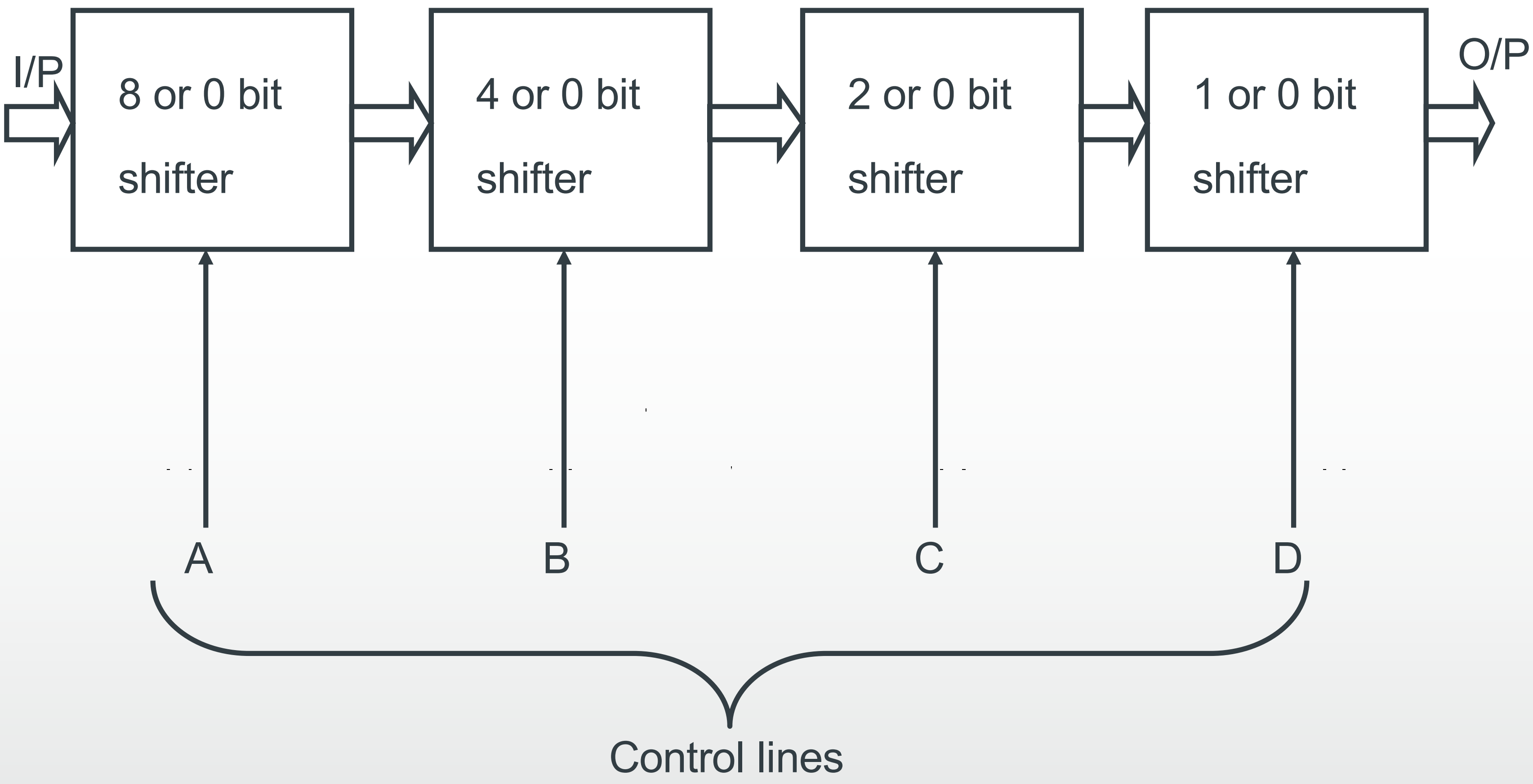
Logarithmic shifter



Shift by 1 place,
then by 2 places

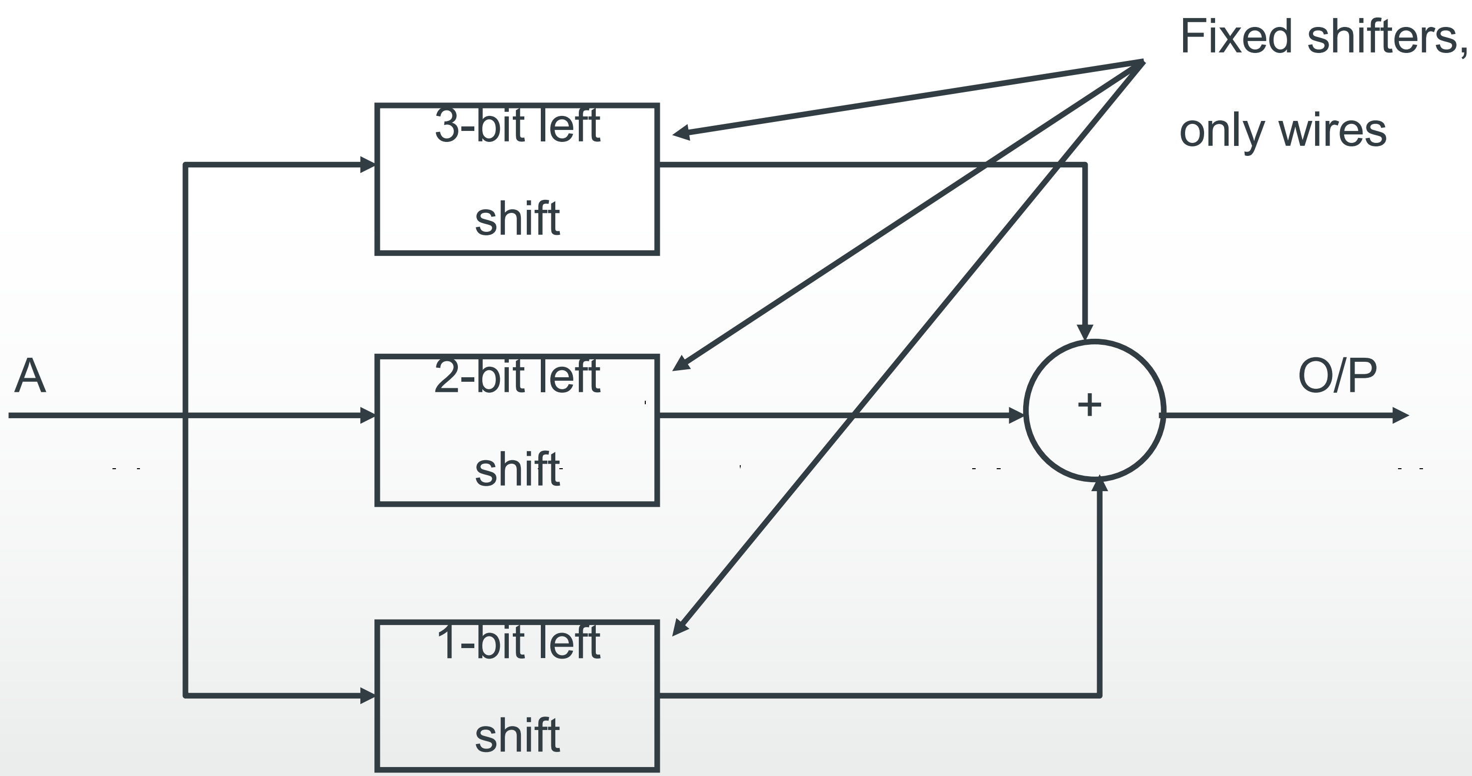
Simplified structure
but more stages
(greater delay)

Programmable shifter example: 16-bit



Multiplication using shifter

$A \times 14 = A\,(1110) = A\,(2^3 + 2^2 + 2^1)$



Shifter: summary

- Trade-offs between area and delay
    - Barrel shifter: fastest O(1), N^2 transistors
    - Logarithmic shifter: O (log N), N log N transistors
    - One bit shifter: O (N), N transistors
- Barrel shifter: wire dominated circuit

•Note. Floating point arithmetic uses shifters a lot to align mantissas

ALUs: Summary

- ALU has to implement
  - Addition (2's complement covers subtraction)
  - Shifting
  - Bitwise logical operations
  - Comparisons can be done by subtraction
- Extended ALU may include
  - Multiplication/Division
- RISC-V allows us to include custom instructions
  - e.g. complex arithmetic

Complex numbers and addition

- Complex numbers have real part and imaginary part

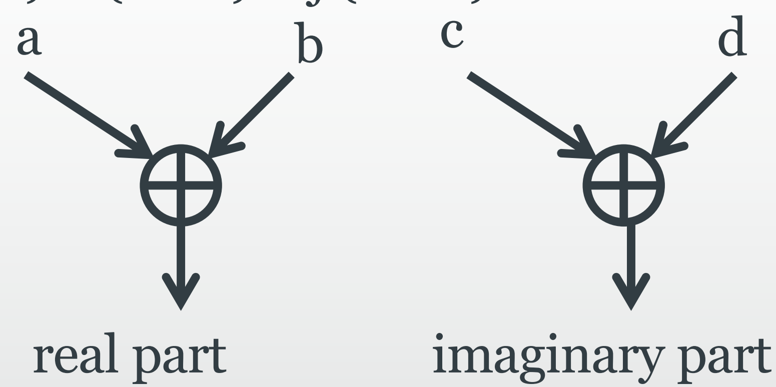
  $x = a + jb$, a = real part, b = imaginary part

  BUT both are just NUMBERS – complex number represents a vector with amplitude $\sqrt{(a^2+b^2)}$ making a phase angle $\theta = \tan^{-1}(b/a)$ with the real axis

  So use two separate bus for them – b-bit complex number = b-bit real number and b-bit imaginary number ≡ 2-b bit wordlength

  But keep the convention, i. e., add real part with real part and imaginary part with imaginary part – don't mix them

  $(a + jb) + (c + jd) = (a + c) + j\,(c + d)$



The complexity of a b-bit complex adder is 2b full adders

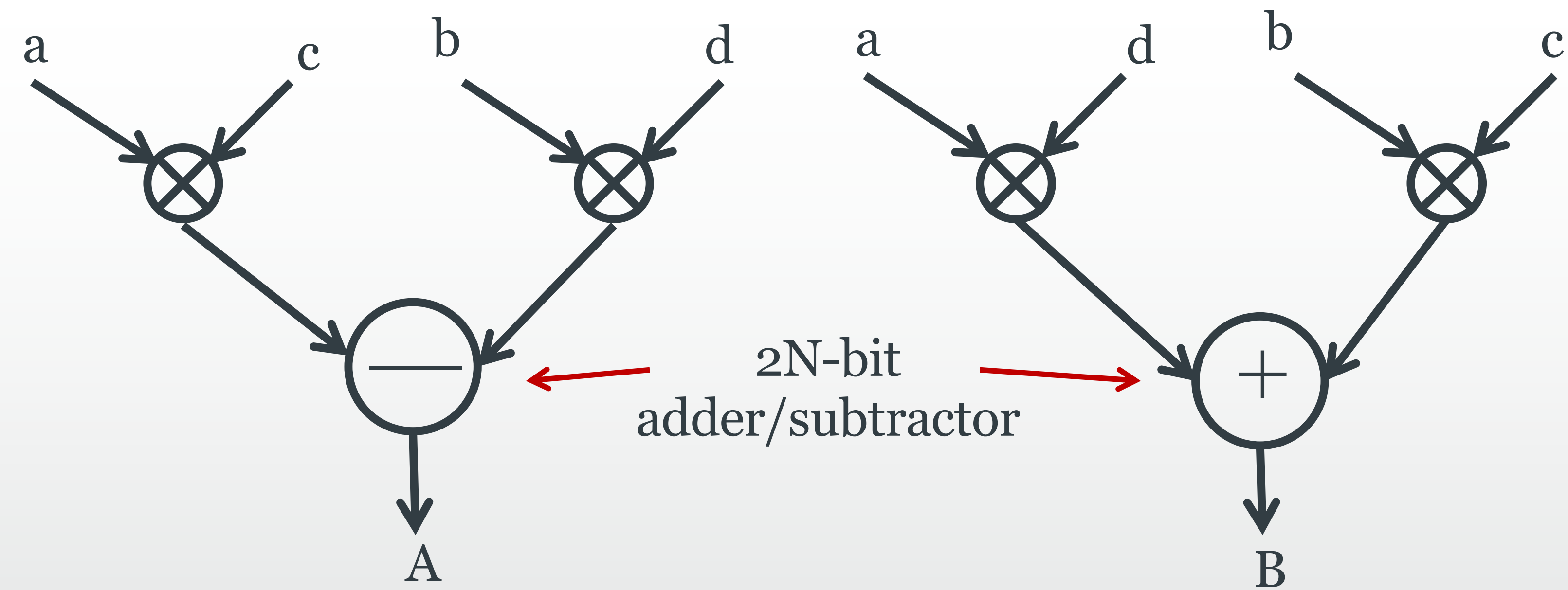
Multiplication of complex numbers

- Addition is straightforward, what about multiplication?

  x = a + jb, y = c + jd

$$xy = (ac - bd) + j(ad + bc) = A + j\,B$$

- So one complex multiplication in reality is 4 real multiplications + two addition/subtraction



Multiplication of complex numbers

- Overall complexity?

  1 N-bit real multiplier = N(N-2) full adders, N half adders and N^2 AND gates

  Total complexity = 4 [N(N-2) full adders + (N/2) full adders + N^2 AND] + 2 (2N-bit) full adders = 2N (2N – 1) full adders + N^2 AND gates

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- Can we reduce it?

  $xy = (ac - bd) + j(ad + bc) = [c(a + b) - b(d + c)] + j\,[d(a - b) + b(c + d)]$

  We can reuse the term b(c + d) resulting in overall complexity of 3 N-bit adders, 3 (N+1)-bit multipliers and 2 (2(N + 1))-bit adders ≈(1/2)[$6N^2$+17N+5] full adders + $3(N+1)^2$ AND gates

Reduced complex multiplier diagram